GAURANSH GOEL

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Education

Columbia University, New York Aug 2026 - Dec 2027
Master of Science, Computer Science

Relevant Courses: Artificial Intelligence, Deep Learning, Computer Vision, Data Structures and Algorithms

Guru Gobind Singh Indraprastha University, New Delhi Nov 2022 - Jul 2026
Bachelor of Technology, Computer Science and Engineering *CGPA - 9.39 / 10*

Relevant Courses: Reinforcement Learning, Machine Learning, Database Management Systems, Software Engineering

Technical Skills

Programming Languages: Java, C, C++, Python, PHP
Frontend and Web: HTML, CSS, JavaScript, TypeScript, ReactJS, NodeJS, Tailwind, Bootstrap, ShadcnUI, Zustand
Framework/Tools: NextJS, Express, TensorFlow, Figma, Git, GitHub
Databases: MongoDB, MySQL, PostgreSQL

Work Experience

Amazon - ML Summer School Trainee Aug 2025

- Selected among the **top 150** participants from **3,000+** applications for this exclusive program.
- Gained hands-on exposure to supervised & unsupervised learning, deep learning, generative AI & LLMs, reinforcement learning, dimensionality reduction, sequential learning, and causal inference through a series of 8 sessions with Amazon scientists.

Delhi Metro Rail Corporation Limited (DMRC) - Software Development Engineer Intern Jun 2025 - Jul 2025

- Developed a full-stack Vendor Empanelment Portal featuring department-specific dashboards, responsive UI, role-based workflows, and automated approval processes, **replacing manual onboarding** with a streamlined digital process.
- Implemented **OAuth**-based authentication, **RESTful APIs**, and role-based access control.
- Led the design of **relational database schemas** with structured **CRUD operations**, integrating backend logic for real-time application status tracking, multi-user communication, and document uploads.

Bharat Electronics Limited (BEL) - Machine Learning Intern Jun 2024 - Jul 2024

- Built and trained a high-precision ML model improving shipping vessel identification and collision avoidance by **60%**.
- Applied data standardization, augmentation, and visualization techniques to address limited and noisy marine data, resulting in a **140%** dataset enhancement.
- Conducted hyperparameter tuning and compared several models to achieve a base precision of **92%**.

Research

Multimodal Hate Speech Detection on Memes using Deep Learning Sep 2024 - May 2026

- Implemented a comprehensive **feature extraction pipeline** using **BERT** for semantic text understanding and Vision Transformer (**ViT**) for image analysis, processing both original and generated captions.
- Engineered a **feature fusion** approach combining visual and textual features to improve **binary classification** accuracy in identifying meme content as hateful or non-hateful.

CNN-Based Weapon Detection Software Across Varied Weather Conditions Oct 2023 - Oct 2024

- Researched and designed a weapon detection system using a **Convolutional Neural Network** model.
- Achieved **85%** detection accuracy by optimizing CNN architecture and data preprocessing techniques.

Projects

Gesture & Voice Virtual Mouse (Python, MediaPipe, OpenCV, Supabase, PyAudio) Feb 2026 - May 2026

- Developed a real-time multimodal HCI (Human-Computer Interaction) system integrating gesture recognition and voice control, enabling touchless cursor navigation, click actions, and drag-and-drop.
- Built web dashboard for real-time monitoring and deployment of system-level voice commands; work published as a research paper.

PoseCheck (MediaPipe, TensorFlow, ReactJS, MongoDB, NodeJS) Aug 2025 - Nov 2025

- Designed multiple exercise-specific posture-detection algorithms (bicep curls, squats, lunges, plank) using angle-based ML logic, enabling real-time form feedback, repetition counting, and session storage; published as a review paper.

CampusPro (NodeJS, PostgreSQL, NextJS, MaterialUI, CSS, AntDesign) Mar 2024 - Jul 2024

- Built a full-stack institute management ERP portal supporting three user roles (student, faculty, non-faculty), featuring an admin-controlled real-time notification service and responsive dashboards.

Leadership and Achievements

- Devised and led a piezoelectric-based sustainability solution, securing **1st place** among **50+** teams.
- Shortlisted among **300+** participants at Smart India Hackathon 2024; built an AI-powered applicant simulation tool for DRDO using natural language processing and computer vision to improve interview evaluation.